
Algorithm 1 Sampling-Based Retargeting with Multi-Scale Contact Reward and Part-wise Penetration Constraints

Require: reference $x_{0:T}^{\text{ref}}$ (qpos, object pose, contact mask m), horizon H , samples N_W , iters N , elite frac ρ , min-valid frac f , temperature λ , anneal β_1, β_2 , tol ϵ

- 1: initialise control knots $U \leftarrow \text{interp}^{-1}(u^{\text{ref}})$
- 2: **for** each simulation step $s = 0, 1, \dots$ **do** ▷ outer receding horizon
- 3: $J_{\text{best}} \leftarrow \infty, i \leftarrow 0$
- 4: **while** $i < N$ **and** not converged **do** ▷ inner CEM
- 5: $i \leftarrow i + 1$; update Σ_h^i by anneal schedule (β_1, β_2)
- 6: **for** $j = 1, \dots, N_W$ **do**
- 7: $[W]_j \sim \mathcal{N}(0, \Sigma^i)$; $[U^i]_j \leftarrow U^i + [W]_j$
- 8: roll out $x_{0:H}^j$ in the physics simulator from $u^j = \text{interp}([U^i]_j, \tau)$
- 9: $J^j, \mathbf{g}^j \leftarrow \text{COSTANDGATE}(x_{0:H}^j, x^{\text{ref}})$ ▷ Alg. 2
- 10: **end for**
- 11: $U^{i+1} \leftarrow \text{GATEDELITE}(\{J^j\}, \{\mathbf{g}^j\}, \{[U^i]_j\}, \rho, f, \lambda)$ ▷ Alg. 2
- 12: **if** $|J_{\text{best}} - \min_j J^j| < \epsilon$ **then**
- 13: **break** ▷ early stop
- 14: **end if**
- 15: $J_{\text{best}} \leftarrow \min_j J^j$
- 16: **end while**
- 17: execute first n_{exec} controls of U ; advance simulator
- 18: **end for**
- 19: **return** refined dynamically-feasible controls U^*

Algorithm 2 Multi-Scale Contact Reward and Part-wise Penetration Constraints, with Gated-Elite Selection

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1: function COSTANDGATE( $x_{0:H}, x^{\text{ref}}$ )
2:    $J \leftarrow 0$ 
3:   for  $t = 0, \dots, H$  do
4:      $J \leftarrow J + \underbrace{\|(q_t - q_t^{\text{ref}}) W_{\text{loc}}\|}_{\text{local-frame tracking}} + \underbrace{\alpha_o \|o_t - o_t^{\text{ref}}\|}_{\text{object pose}}$ 
5:     for each end-effector  $e$  with contact  $m_t^e = 1$  do  $\triangleright$  C1: Multi-Scale Contact Reward
6:        $r_{t,e}^{\text{coarse}} \leftarrow g \exp(-\|p_t^e - p_t^{\star,e}\|/\sigma_c)$   $\triangleright$  broad scale ( $\sigma_c$ )
7:        $r_{t,e}^{\text{fine}} \leftarrow s \exp(-|\phi_e(x_t)|/\sigma_f) \cdot \mathbf{1}[\phi_e(x_t) \in B]$   $\triangleright$  sharp scale ( $\sigma_f$ )
8:        $J \leftarrow J - r_{t,e}^{\text{coarse}} + r_{t,e}^{\text{fine}}$ 
9:     end for
10:    for each part group  $P \in \{\text{hand, body, leg}\}$  do  $\triangleright$  C2: Part-wise Penetration
11:       $J \leftarrow J + w_P \max(0, \delta_P - \phi_P(x_t))$ 
12:    end for
13:     $J \leftarrow J + \text{ground-penetration terms (body / foot height } z)$ 
14:  end for
15:  for each part group  $P$  do  $\triangleright$  hard
16:     $g_P^{\min} \leftarrow \min_t \phi_P(x_t); \quad g_P^{\text{vio}} \leftarrow \frac{1}{H} \sum_t \mathbf{1}[\phi_P(x_t) < \theta_P]$ 
17:  end for
18:  return  $J, \mathbf{g} = \{(g_P^{\min}, g_P^{\text{vio}})\}_P$ 
19: end function

20: function GATEDELITE( $\{J^j\}, \{\mathbf{g}^j\}, \{U_j\}, \rho, f, \lambda$ )
21:    $\text{valid}_j \leftarrow \bigwedge_P [g_P^{\min,j} \geq \text{floor}_P \wedge g_P^{\text{vio},j} \leq \tau_P]$   $\triangleright$  per-part AND
22:   if  $\#\{j : \text{valid}_j\} \geq f N_W$  then
23:      $\mathcal{E} \leftarrow \text{top-}\lceil \rho N_W \rceil \text{ of } \{j : \text{valid}_j\} \text{ by lowest } J^j$ 
24:   else  $\triangleright$  least-violation fallback
25:      $\mathcal{E} \leftarrow \text{top-}\lceil \rho N_W \rceil \text{ by lowest } (\text{depth}^j + \text{vio}^j - 10^{-3} \tilde{J}^j)$ 
26:   end if
27:    $w_j \leftarrow \frac{\exp(-J^j/\lambda) \mathbf{1}[j \in \mathcal{E}]}{\sum_{k \in \mathcal{E}} \exp(-J^k/\lambda)}$ 
28:   return  $\sum_j w_j U_j$   $\triangleright$  softmax elite refit
29: end function

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